

IncidentIQ+: Enterprise Agentic Incident Response System

Overview

IncidentIQ+ is an enterprise-grade AI system designed for large-scale infrastructure environments. It leverages NemoClaw for agentic workflows, vLLM for high-performance inference, and introduces governance, simulation, and speculative reasoning to safely automate incident response.

Track Alignment

Track 5 – Agentic Edge powered by NemoClaw (Builder / Deep Tech)

Problem

- Massive log volumes in distributed systems
- Slow manual debugging (30–120 mins)
- Fragmented knowledge across tools and docs
- High risk in executing production fixes

Solution

IncidentIQ+ transforms raw infrastructure data into intelligent decisions and controlled actions.

- Log understanding
- Root cause reasoning
- Policy-based decision making
- Tool execution & simulation
- Enterprise-safe automation

Core Innovations

1. Tiered Governance (Steerability)

- Tier 1: Autonomous fixes (restart, retry)
- Tier 2: High-risk actions (scaling, config changes)
- Risk scoring before execution
- NemoClaw policy enforcement

2. Speculative Agentic Loop (Inference Optimization)

- Small model drafts commands
- Large model verifies (vLLM)
- Reduces latency and improves accuracy

3. Terminal-Based Agent Interface

- CLI-based workflow
- Streams reasoning and tool calls
- Aligns with real DevOps usage

4. Shadow Simulation Mode

- Test fixes in sandbox container
- Validate logs before production
- Ensures safe execution



Architecture

User Input → NemoClaw Agent → Policy Check → Tool Calling → Speculative Loop → Shadow Simulation → Final Action



Tools

- Log Analyzer
- Root Cause Detector
- Severity Classifier
- Command Generator
- Risk Scoring Engine
- Simulation Engine
- Incident Report Generator



Users

- DevOps Engineers
- SREs
- Engineering Managers



Example Flow

- Input: Pod crash (OOMKilled)
- Analyze logs
- Detect root cause
- Classify severity
- Generate fix
- Run speculative verification

- Simulate in shadow pod
- Output final safe action

Output

- Severity: Critical
- Cause: Memory Limit Exceeded
- Fix: Increase memory
- Risk Score: Medium
- Simulation: Passed
- Status: Safe to Execute

Impact

- 40–90% faster resolution
- Reduced downtime
- Safer automation
- Enterprise-ready AI workflows

Final Pitch

IncidentIQ+ is a NemoClaw-powered enterprise agent that diagnoses, governs, and safely executes infrastructure fixes using speculative reasoning and simulation—bringing real autonomous operations to life.